DESHMUKH TECHNOLOGIES

DTTP · V1.0

Software Setup & Environment

Trainee Onboarding — Windows Verification Guide

Every trainee independently installs, configures, verifies, clones, builds, runs, tests, commits, and pushes before assigned project work.

DOCUMENT TYPE Trainee Onboarding Standard	PLATFORM Windows 10 / 11
PRIMARY STACK Java · Spring Boot · React · MySQL · Git · Docker	MOBILE TRACK Flutter + Android Studio
ALTERNATE BACKEND Python / FastAPI when assigned	AUDIENCE Trainees, mentors, coordinators, admin

Install. Configure. Verify. Then build.

INTERNAL TALENT DEVELOPMENT STANDARD

DOCUMENT	VERSION	CLASSIFICATION	OWNER
DTTP Software Setup	1.0	Internal Onboarding Standard	Deshmukh Technologies

GUIDE

Contents

PART I Purpose & machine

01 Purpose

This document is the standard software and development environment for every Deshmukh Technologies trainee. Day 1 is not a lecture day. It is the day the machine becomes a working engineering station.

INSTALL CONFIGURE VERIFY CLONE BUILD RUN TEST COMMIT PUSH

Document	Role
This setup guide	Machine, tools, GitHub, Day-1 sign-off
90-Day DTTP Handbook	Timed training path, project, assessment
Self-Growth Program	Ten-level knowledge map

> Complete this path before assigned project development. Employee Management features do not start until the mentor marks the environment Pass or Conditional.

02 Scope

Installation, Windows layout, Git, GitHub SSH, Java / Spring Boot, Node / React, MySQL, Postman, Docker, Flutter for the mobile track, verification, troubleshooting, trainee sign-off, mentor verification, and Day-1 completion. Python / FastAPI is installed only when that backend track is assigned. JDK remains the DTTP default.

03 Standard software checklist

Software	Purpose	Required
Windows 10 / 11	Operating system	Yes
Git + GitHub account	Version control and source	Yes
VS Code	General / frontend	Yes
IntelliJ IDEA	Java / Spring Boot	Yes
JDK 17 (or 21 if the project says so)	Java	Yes
Maven	Java build	Yes
Node.js LTS + npm	React	Yes
MySQL Server + Workbench	Database	Yes
Postman + Chrome	API and debugging	Yes
Docker Desktop	Containers	Recommended
Flutter SDK + Android Studio	Mobile	Mobile track
Python 3.11+	FastAPI	Python track only

04 Hardware requirements

MINIMUM

- 8 GB RAM
- 256 GB SSD
- Modern dual / quad-core CPU
- Stable internet
- Windows 10 / 11

RECOMMENDED

- 16 GB RAM
- 512 GB SSD
- Intel Core i5 / AMD Ryzen 5 or better
- Full HD display
- Reliable broadband

A machine below the minimum will struggle with IntelliJ + Chrome + Docker + MySQL at once. Tell the mentor before Day 1 if the machine is below minimum.

05 Windows preparation

Update: Settings → Windows Update → Check for updates. Install security and system updates.

Restart after major updates. Do not install JDK, Docker, or Android Studio on a pending-reboot machine.

Create `C:\Development\` with `Projects`, `Tools`, `Documents`, and `Backups`.

> Do not keep clones in C:\Windows, C:\Program Files, Desktop, or Downloads.

Recommended: enable Windows long paths from an Administrator PowerShell so deep Node and Java trees do not fail.

```
New-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Control\FileSystem" `
  -Name "LongPathsEnabled" -Value 1 -PropertyType DWORD -Force
```

PART II Tools

06 Git installation

Install Git for Windows. Enable Git from the command line, bundled OpenSSH, default branch `main`. Open a **new** PowerShell:

```
git --version
git config --global user.name "Your Name"
git config --global user.email "your-email@example.com"
git config --global init.defaultBranch main
git config --global --list
```

Expected: `git version 2.x.x`. Use the email on the GitHub profile.

07 GitHub account

Each trainee has an authorized GitHub account and repository access from the mentor or admin. Never share passwords, personal access tokens, private SSH keys, or company secrets. Access is granted to named people, not shared logins.

08 GitHub SSH configuration

```
ssh-keygen -t ed25519 -C "your-email@example.com"
Get-Service ssh-agent | Set-Service -StartupType Automatic
Start-Service ssh-agent
ssh-add $env:USERPROFILE\.ssh\id_ed25519
Get-Content $env:USERPROFILE\.ssh\id_ed25519.pub
```

Accept the default key path. Set a passphrase. If `Set-Service` is denied, run those two service lines as Administrator, then `ssh-add` in a normal window.

Copy the **public** key only (starts with `ssh-ed25519`, one line). GitHub → Settings → SSH and GPG keys → New SSH key. Title: `DTTP-Windows-<name>`.

```
ssh -T git@github.com
Hi <github-username>! You've successfully authenticated, but GitHub does not provide shell access.
```

First connection may ask to trust GitHub's host key. Type `yes`. Never copy `id_ed25519` without `.pub`.

09 VS Code

Prettier | ESLint | GitLens | EditorConfig | Docker | REST Client | Java pack | Spring Boot pack

Format On Save: On. Auto Save: afterDelay or onFocusChange. Default formatter: Prettier for JavaScript, TypeScript, and JSON. React snippet packs are optional; prefer typing the code in training.

10 IntelliJ IDEA

Use IntelliJ for Java, Spring Boot, Maven, JUnit, Mockito, JPA, and Spring Security. Community is enough unless the company issues Ultimate.

Setting	Value
Project SDK	JDK 17 (or 21 if the project says so)
Build tool	Maven
Version control	Git
Annotation processing	Enabled for Spring projects

Do not mix VS Code and IntelliJ on the same Java change in the same hour.

11 Java JDK

DTTP default: **JDK 17 LTS**. Install a full JDK, not a JRE.

```
JAVA_HOME = C:\Program Files\Java\jdk-17
Path      += %JAVA_HOME%\bin
```

Point `JAVA_HOME` at the JDK root, not at `bin`. Folder names vary by vendor (Temurin, Microsoft, Oracle). New terminal:

```
java -version
javac -version
echo $env:JAVA_HOME
```

Both commands must work. `java` without `javac` means a JRE is on PATH. Expected: `openjdk version "17.0.x"` and `javac 17.0.x`.

12 Maven

Install Maven 3.9.x or use the project wrapper. Prefer the wrapper when `mvnw.cmd` exists:

```
mvn -version
.\mvnw.cmd clean install
```

Do not upgrade Maven globally to fix a wrapper build.

13 Spring Boot verification

```
cd C:\Development\Projects
git clone git@github.com:<org>/<backend-repo>.git
cd <backend-repo>
.\mvnw.cmd clean install
.\mvnw.cmd spring-boot:run
```

Copy local config from the example. Never commit it.

```
spring.datasource.url=jdbc:mysql://localhost:3306/dttp_training
spring.datasource.username=dttp_dev
spring.datasource.password=use-a-local-password
```

Pass when: `BUILD SUCCESS`, process stays up, logs show MySQL connected, `GET http://localhost:8080/api/health` returns `{ "status": "ok" }` (or the project health path).

14 Node.js

Install the Node.js LTS that matches the frontend (`package.json` engines and `.nvmrc` if present). Expected: Node 20.x or 22.x unless the project pins another version.

```
node --version
npm --version
```

Do not change Node versions to "fix" `npm install` on an existing lockfile.

15 React environment

```
cd C:\Development\Projects
git clone git@github.com:<org>/<frontend-repo>.git
cd <frontend-repo>
copy .env.example .env
npm install
npm run dev
```

Some projects use `npm start`. Follow `package.json`. Chrome must load the printed URL (often `http://localhost:5173`). Be able to point at components, props, state, hooks, routing, service-layer API calls, forms, errors, and env vars. Never commit secrets.

16 MySQL installation

Install MySQL Server 8.x and Workbench. Local root password in a password manager, not in chat. Windows service enabled. Port **3306**.

Field	Value
Host	localhost
Port	3306
Training database	`dttp_training`

Never use production credentials on a trainee machine.

17 MySQL verification

Connect in Workbench. The trainee must create a database, create a table, insert, query, update, and delete.

```
CREATE DATABASE IF NOT EXISTS dttp_training
  CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER IF NOT EXISTS 'dttp_dev'@'localhost' IDENTIFIED BY 'use-a-local-password';
GRANT ALL PRIVILEGES ON dttp_training.* TO 'dttp_dev'@'localhost';
USE dttp_training;
CREATE TABLE smoke_test (id INT PRIMARY KEY AUTO_INCREMENT, label VARCHAR(80) NOT NULL);
INSERT INTO smoke_test (label) VALUES ('DTTP Day 1');
SELECT * FROM smoke_test;
```

18 Postman

Learn GET, POST, PUT, PATCH, DELETE, URL, query, path, headers, JSON, Bearer token, and status codes.

POST /API/AUTH/LOGIN | JWT | AUTHORIZATION BEARER | GET /API/EMPLOYEES | GET /API/HEALTH

Save the collection under `C:\Development\Documents`. No production tokens.

19 Docker Desktop

WSL 2 backend required. Enable virtualization if Docker reports it off.

```
docker --version
docker compose version
docker run hello-world
```

Understand image, container, Dockerfile, volume, network, Compose. On 8 GB RAM, start Docker only when the day's work needs it. Mentor may waive Docker in writing.

20 Flutter — mobile track

Install Flutter SDK (stable), Android Studio, Android SDK, emulator or authorized device with USB debugging. Add Flutter `bin` to PATH.

```
flutter --version
flutter doctor
flutter doctor -v
cd C:\Development\Projects
flutter create dttp_test_app
cd dttp_test_app
flutter run
```

Pass when the counter app launches. Resolve every red `flutter doctor` item on this track.

21 Chrome DevTools

Press **F12**. Elements (HTML/CSS), Console (JS errors), Network (API status and payload), Application (storage and cookies), Sources (breakpoints). Day-1: load the React app and confirm the document request, plus one XHR/fetch when the API is up.

PART III Verify

22 Environment verification

Run in a **new** PowerShell. Paste output into the sign-off.

```
git --version
java -version
javac -version
mvn -version
node --version
npm --version
mysql --version
docker --version
docker compose version
ssh -T git@github.com
```

Mobile: `flutter --version` and `flutter doctor`. Python track: `python --version` (3.11 or 3.12) and `pip --version`. Skip Docker commands only with a written mentor waiver.

23 Project verification

01 **GitHub repository**	02 **Clone with SSH**
03 **Install dependencies**	04 **Configure local env (not committed)**
05 **Database connection**	06 **Build**
07 **Run backend**	08 **Run frontend**
09 **Test API**	10 **Small allowed change**
11 **Commit**	12 **Push**

First commit only after the mentor names a safe file:

```
git checkout -b dttp/<initials>/env-setup
git add docs/setup-notes.md
git commit -m "docs: record Day-1 environment verification"
git push -u origin dttp/<initials>/env-setup
```

Do not commit `.env` or `application-local.properties`.

24 Environment variables

Never hard-code `DB_PASSWORD`, `JWT_SECRET`, `API_KEY`, or AWS keys. Use `.env`, `application-local.properties`, or `application-dev.properties`. Confirm secrets are in `.gitignore`.

25 Recommended .gitignore

```
.env
.env.local
application-local.properties
.idea/
.vscode/
node_modules/
build/
dist/
target/
*.log
*.jks
*.pem
*.p12
id_ed25519
```

If a secret was committed by mistake, tell the mentor immediately. Do not only delete the file in a later commit.

26 Troubleshooting

Symptom	First check
Git / Java / Node not recognized	Install, PATH, **new** PowerShell. JDK needs `javac`, not only `java`.
Maven not recognized	Use `.\mvnw.cmd` in the project.
`npm install` fails	Match Node to `engines` / `.nvmrc`. Do not change lockfile versions at random.
MySQL connection fails	Service running, port 3306, user, password, database exists, host is `localhost`.
Docker does not start	Desktop running, virtualization, WSL 2, `docker info`.
SSH asks for a GitHub password	Remote is HTTPS. Set origin to `git@github.com:<org>/<repo>.git`.
Flutter doctor red	`flutter doctor -v`. Fix SDK, licenses, emulator, or PATH first.

27 Security rules

- [ ] Never share passwords on WhatsApp or email
- [ ] Never commit credentials to GitHub
- [ ] Never expose API keys
- [ ] Never share private SSH keys
- [ ] Never upload company source to unauthorized locations
- [ ] Never use production credentials for training
- [ ] Never install unauthorized software on a company machine
- [ ] Report security incidents immediately

PART IV Sign-off

28 Trainee setup checklist

WINDOWS & TOOLS

- [ ] Windows updated and restarted
- [ ] C:\Development folders created
- [ ] Git identity configured
- [ ] VS Code + IntelliJ installed
- [ ] java and javac both work
- [ ] Maven wrapper or mvn verified
- [ ] Node LTS and npm verified
- [ ] Chrome installed

DATA, GITHUB, TRACKS

- [ ] MySQL 3306 + dttp_training smoke SQL
- [ ] Postman collection saved
- [ ] Docker hello-world or written waiver
- [ ] SSH public key on GitHub
- [ ] ssh -T git@github.com succeeded
- [ ] Assigned repo cloned with SSH
- [ ] Mobile: flutter doctor + test app
- [ ] Python track: python 3.11+ if assigned

29 Python / FastAPI track

Install only when the mentor assigns this track. Same MySQL database, Git, and security rules. Health remains `GET /api/health`.

```
python --version
pip --version
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
uvicorn app.main:app --reload
```

If activation is blocked: `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`.

30 Evidence pack

Attach screenshots or pasted logs to the Day-1 report. Crop secrets.

Evidence	Shows
`git --version` and global config	Git identity
`ssh -T git@github.com`	SSH works
`java` / `javac` and Node / npm	JDK and Node
Workbench on `dttp_training`	Database
Backend BUILD SUCCESS or run log	Spring Boot
Browser with the React app	Frontend
Postman `GET /api/health`	API
GitHub setup branch	Push
`flutter doctor` (mobile)	Mobile track

31 Trainee sign-off

A verbal “it’s done” is not Day-1 complete.

Field	Trainee fills
Full name	
Date	
GitHub username	
Machine (make / RAM / disk)	
Windows version	
Backend track (Java / Python)	
Mobile track (Yes / No)	
Docker (Installed / Waived)	
Assigned repos	

I confirm that the required tools are installed, SSH works, I can clone, build, and run the assigned starter, no secrets are in Git, and the evidence pack is attached.

Trainee signature / typed name

Date

---

32 Mentor / admin verification

Spot-check: `ssh -T`, `java` and `javac` (or Python), MySQL `dttp_training`, one running app, `.gitignore` covers `.env`, remote is `git@github.com:` not HTTPS.

Result	Meaning
Pass	Day-1 complete. Trainee may start Phase 1 technical work.
Conditional	Listed gaps with a fix date within 48 hours.
Fail	Environment is not usable. Repeat setup with mentor support.

Mentor name

Result — Pass / Conditional / Fail

Notes

Date

---

33 Day-1 completion criteria

- [ ] Core verification commands succeed
- [ ] GitHub SSH authentication succeeds
- [ ] Assigned repo cloned into C:\Development\Projects
- [ ] dttp_training exists and smoke SQL ran
- [ ] Backend or frontend starter runs locally
- [ ] One API call shown in Postman or Network
- [ ] One allowed commit pushed on a feature branch
- [ ] Trainee sign-off filled
- [ ] Mentor result is Pass or Conditional with dated gaps

> Until Day 1 is complete, the trainee does not start Employee Management features.

DTTP — DESHMUKH TECHNOLOGIES TRAINEE PROGRAM

Install. Configure. Verify. Then build.

Windows station ready. GitHub authenticated. Database local. First commit on a branch.

Learn. Build. Solve. Review. Deploy. Grow.